

Builder Basics — The Living Village (Extension)

Topic: Bring It All Together · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 90 minutes

🌱 This is a bonus challenge, not a weekly lesson. Try it after **Complete Your Village**, once you can build with loops, move the agent between spots, and place water and animals.

1 · The Challenge

You already finished a small village. Make one that feels **alive** — crops growing, animals wandering, water you can cross.

By the end, your village must have **all eight** of these:

 A **farm** — rows of crops  A **farm house** — where the farmer lives  A **house with a watchtower** — a home with a tall tower on top  A **fence** — around the farm or the animals

 A **pond with fish** — water with a fish in it  A **bridge** — one that crosses over the pond

 A **well** — for water  A **few animals** — cows, sheep, pigs, or chickens

There is no fixed size and no map — **you** decide how big each thing is and where it goes. One rule you must not break:

🚧 **Everything you can build must be built with code.** No placing blocks by hand. (Animals are spawned — code or a spawn egg is fine.)

List the eight things your village must have when you finish.

The last village was built with code. Why does building with code get *easier* the bigger your village gets?

2 · Plan Before You Build

A builder plans first. For **each** part, decide the blocks, the size, and whether a loop can do the repeating.

Farm — What crop? How many rows, and how long is each row?

Farm house — What blocks for the walls and roof? How wide and how tall?

House with a watchtower — Where does the tower sit — a corner, the middle, the roof? How much taller than the house is it?

Fence — What does it go around? How will the agent turn the corners?

Pond with fish — How will you make the dip in the ground? What fills it, and where do the fish go?

Bridge over the pond — How wide is the pond at its narrowest? How long must the bridge be to reach across?

Well — What shape is the hole? What goes around the top? Is there water inside?

Animals — Which animals, and how many? Will they go inside the fence?

3 · Think in Loops

Eight structures is a lot to place one block at a time. A loop is for anything you do **again and again** — a row of crops, the four sides of a fence, the planks of a bridge, the levels of a tower.

Pick three parts that will use a loop. For each one, say what the loop repeats each time.

Between one structure and the next, the agent has to **walk to a new spot** so nothing lands on top of the last build.

After you build the farm house, how does the agent get to a clean spot to start the watchtower house?

The bridge and the fence are special: they change direction. A fence goes across, then **turns**, then goes across again.

What has to happen to the agent between one side of the fence and the next?

4 · Show Your Work 📷 🎥

Go to your homework world and build each part one at a time, moving the agent between them. Build the water first, then put the bridge across it so you can see it reach.

Keep going until your village has **all eight** parts — the farm, the farm house, the watchtower house, the fence, the pond with fish, the bridge, the well, and the animals.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.